

1. Consider two frames S and S' , with S' travelling with a velocity v wrt S . An object is moving with velocity u' wrt S' . Show that the object will travel with velocity v wrt S where

$$v = \frac{v + u'}{1 + vu'/c^2}$$

2. Consider two frames S and S' with coordinate axes (x, y, z) and (x', y', z') respectively. Let S' move along the x direction with a velocity v . Consider an object moving with velocity u_y in the stationary frame. Show that the object's velocity in S' is

$$u'_x = -v$$

$$u'_y = u_y \sqrt{1 - v^2/c^2}$$

3. Using the Lorentz transformations, show that the time of a moving observer with a velocity u is dilated by a factor of $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$ wrt the stationary observer.
4. Using the Lorentz transformations, show that an object with proper length L will shrink by a factor of $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$ if it moves with a velocity v wrt to a stationary observer.
5. Show that the Lorentz transformations preserve the spacetime interval

$$\Delta s^2 = -c^2 \Delta t^2 + \Delta x^2$$

6. Among the particles of high-energy physics are charged pions, particles of mass between that of the electron and the proton and of positive or negative electronic charge. They can be produced by bombarding a suitable target in an accelerator with high energy protons, the pions leaving the target with speeds close to that of light. It is found that pions are radioactive and when they are brought to rest, their half-life is measured to be 1.77×10^{-8} secs. A collimated pion beam, leaving the accelerated target at a speed of $0.99c$, is found to drop to half its original intensity 39 m from the target.
 - (a) Are these results consistent?
 - (b) Show how the time dilation accounts for the measurement.
 - (c) Show how the length contraction accounts for the measurement.

7. Two spaceships, each measuring 100m in its own rest frame, pass by each other travelling in opposite directions. Instruments on spaceship A determine that the front end of spaceship B requires 5×10^{-6} sec to traverse the full length of A.
 - (a) What is the relative velocity of the two spaceships?
 - (b) A clock at the front end of B reads exactly one o'clock as it passes by the front end of A. What will the clock read as it passes by the rear end of A?
8. Imagine a source of light emitting radiation uniformly in all directions in rest-frame S' . Find the distribution of radiation in the S frame in which the source moves at speed $4/5c$. Draw a polar plot. Can you guess why this phenomenon is referred to as the **headlight effect**?
9. A star A is located 16 light-years away from Earth. Suppose that we send a spacecraft with velocity $v = \frac{4}{5}c$ to investigate a recently discovered exoplanet in this star system.
 - (a) According to the spacecraft's occupants, how long does this trip take?
 - (b) According to the spacecraft's occupants, how far did they travel?
10. Two spaceships A and B are approaching each other along perpendicular directions as seen from Earth. If A is observed to have velocity $u_y = -0.9c$ and B is observed to have velocity $u_x = +0.9c$, determine the speed of spaceship A as measured by an occupant of spaceship B.
11. An observer O at rest midway between two sources of light at $x = 0m$ and $x = 10m$ observes the two sources to flash simultaneously. For the second observer O' who is moving at a constant speed parallel to x-axis, one source flashes $13ns$ before the other. Find the speed of O' relative to O in units of c .
12. Two spaceships approach Earth with equal speeds (as measured by an observer on Earth), but in opposite directions. A meter stick on one spaceship is measured to be $60cm$ long by an occupant on the other spaceship. What is the speed of each spaceship as measured from Earth in units of c ?
13. The classic ladder in a barn paradox posits that a ladder with length l moves with a velocity v with respect to a stationary barn of length $l_B < l$ with doors at its entrance and exit. In the barn's frame, the ladder will be length contracted, and the ladder can fit into the barn, allowing the doors to be momentarily closed with the ladder completely inside. The doors are subsequently opened allowing the ladder to leave. In the ladder's frame of reference, the barn's length is contracted. Explain the apparent contradiction.

14. In the twin paradox, one of the twins travels to a nearby celestial body and returns back to earth younger, relative to his twin. Show that the two twins' path in spacetime is not symmetric wrt each other i.e the moving twin does not belong to a single inertial frame of reference throughout the trip. For simplicity, assume that the moving twin instantly changes direction when returning back.

15. Relativistic Doppler Shift

- (a) How fast and in what direction must a galaxy be moving if an absorption line found at 550nm (green) for a stationary galaxy is shifted to 450nm (blue-shift)?
- (b) How fast is it moving if the same absorption line is shifted to 700nm (red-shift)?